

BESS OPERATIONS REPORT

March 2026

Site:	Tashkent BESS
Reporting period:	01 March 2026 — 01 April 2026
Blocks monitored:	70 blocks
Report generated:	2026-05-17 08:45
Report type:	Monthly Operations & Performance Summary

Report generated automatically from SCADA exports. Per-block KPIs use the validated availability engine. RTE / EFC / throughput sourced from cumulative energy counters.

1. Project Details

Parameter	Value
Project name	Tashkent BESS
Reporting period	01 March 2026 — 01 April 2026
Number of blocks	70 blocks
Site	Tashkent BESS
Project Capacity	70 blocks × ~11 MWh (operational 5-95% SOC)
Configuration	70 blocks × 2 containers × multiple racks per container
OEM	Sungrow Power Supply Co. Ltd.

2. Summary

During **01 March 2026 — 01 April 2026**, the Tashkent BESS fleet (70 blocks) delivered **30,867.7 MWh** of discharge energy on **33,179.3 MWh** of charge energy. Site RTE was **93.03%** (above the 85% target). Container-level availability was **97.14%** (above 95% target). Average SOC: **47.11%**. Average SOH (end-of-month snapshot): **99.18%**. **3067** equivalent full cycles completed (avg 43.8 per block). **2.3** day(s) per block (fleet average) flagged as EXCLUDED and removed from RTE / availability calculations.

3. Services Provision

3.1 Preventative Maintenance (PM)

No PM activities in the reporting period.

3.2 Corrective Maintenance

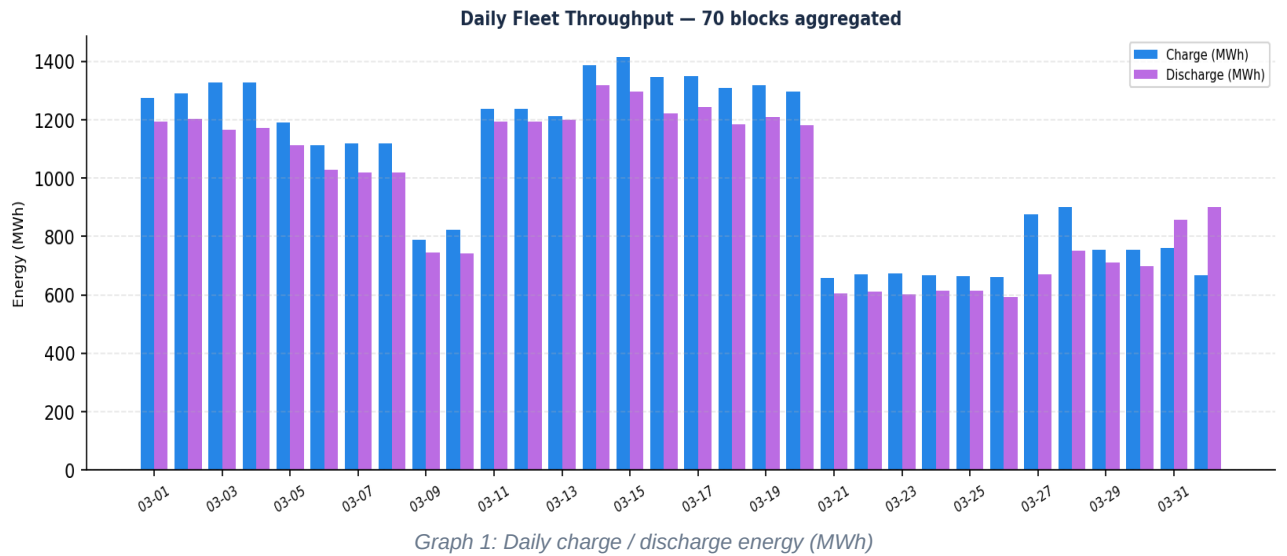
Automated summary based on alarm classification — recurring corrective items observed this period:

Item	Occurrences	Resolution applied
Dry Node Input Fault	185	Check field signal wiring (BMS input)
Fire Alarm (Firefighting)	104	Activate emergency response
BSC-PCS Communication	94	Check ethernet/CAN between BSC and PCS
CMU Fault	29	Check cell monitoring unit
Insulation Impedance Low	28	Insulation resistance test required
DC-DC Converter Fault	12	Restart DC-DC; investigate insulation
Voltage Sample Fault	9	Replace voltage sensor; check wiring

4. Plant's Performance

4.1 Plant Performance Data

No	Item	Total (MWh)
1	Imported Energy (Charge)	33,179.27
2	Exported Energy (Discharge)	30,867.66



Number of Cycles

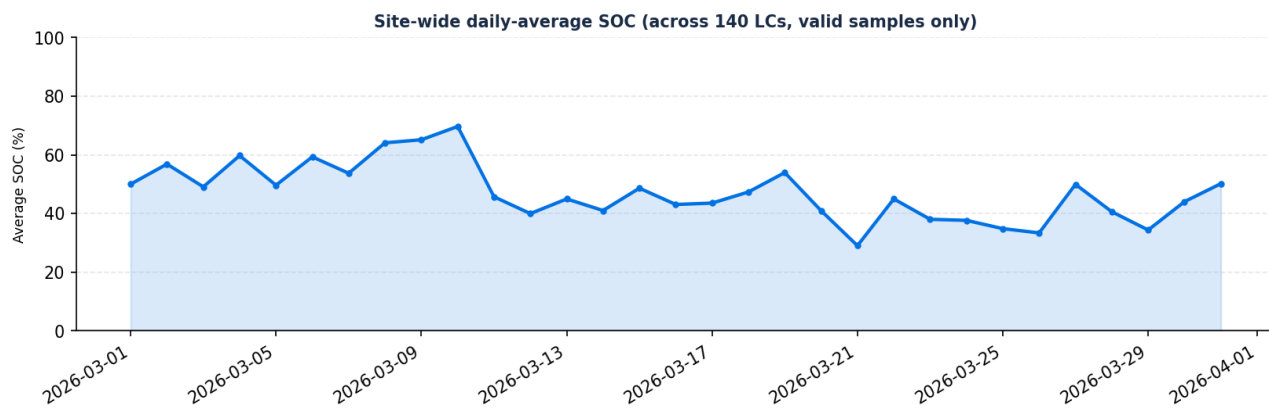
No	Item	Total	% of yearly target (600)
1	Number of Cycles in Reported Month (avg)	43.8	7.30%
2	Accumulative Number of Cycles	43.8	7.30%

Availability Hours (plant-level redundancy method)

Item	Value
Scheduled hours	744.0 h
Scheduled unavailability (PM)	0.0 h
Unscheduled outage (container-aggregate)	5,778.4 h (776.67%)
Plant-level outage (redundancy adjusted)	n/a (plant capacity not configured)
Container-level availability (current engine)	97.14%

4.2 Key Performance Indicators (KPIs)

47.11% Average SOC	99.18% Average SOH (snapshot)	93.03% RTE
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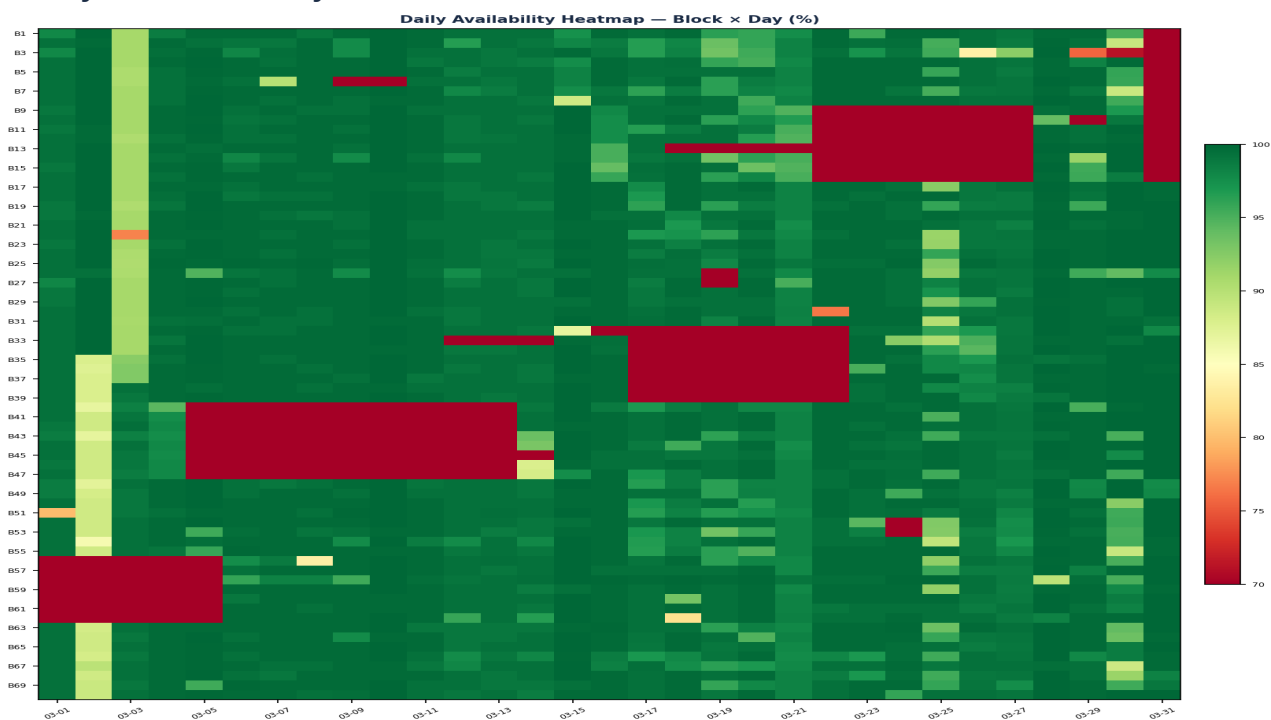


Graph 2: Site-wide average SOC trend (daily)

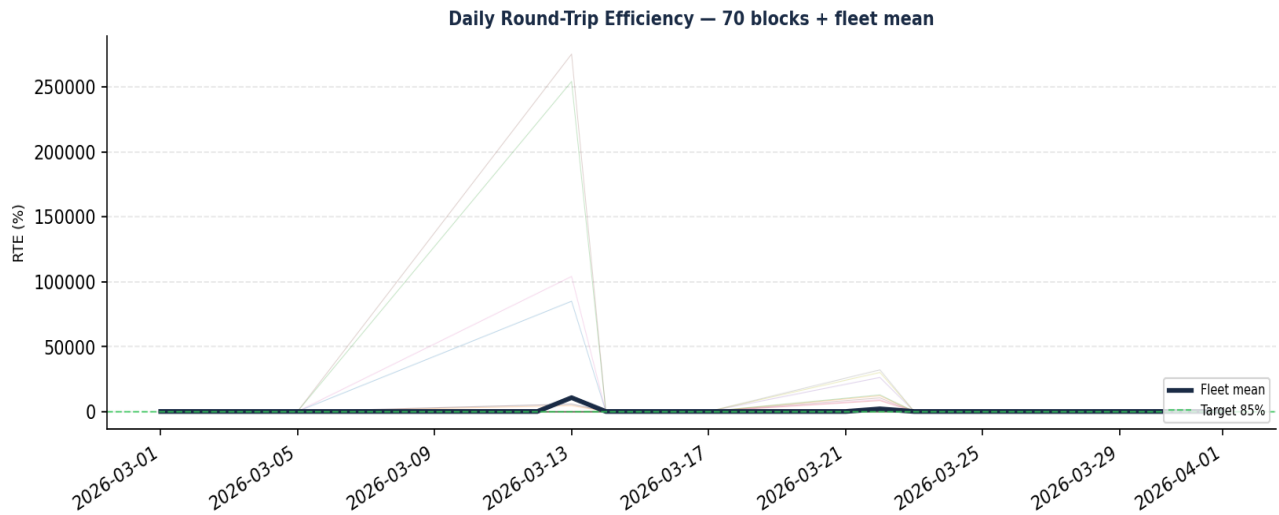
4.3 Monthly Performance Comparison

Month	SOC	SOH	RTE	Cycles	Discharge (MWh)	Charge (MWh)
December 2025	41.0%	99.0%	90.0%	—	2,843	3,146
January 2026	56.0%	99.0%	90.0%	—	6,441	7,122
February 2026	48.0%	99.0%	90.0%	—	5,728	6,336
March 2026	58.0%	99.0%	93.0%	—	7,462	8,068
April 2026	48.3%	97.9%	92.3%	—	6,929	7,509
March 2026	47.1%	99.2%	93.0%	43.8	30,868	33,179

4.4 System Availability



Graph 3: Daily availability heatmap, all 70 blocks. Red = outage, green = full operation. EXCLUDED days removed from calculation.



Graph 4: Daily RTE — light lines = individual blocks, dark line = fleet mean. Target: 85%.

Block performance anomalies (19 block(s) flagged as more than 1 σ below fleet mean):

Block	Avg RTE	RTE z-score	Avg Throughput	Avg Availability	Total Alarms
Block 58	89.64%	-3.37	30.0 MWh	98.57%	8
Block 45	90.23%	-2.00	30.2 MWh	96.20%	0
Block 41	90.25%	-1.98	30.5 MWh	96.33%	0
Block 26	90.34%	-1.76	31.5 MWh	96.65%	3
Block 3	90.38%	-1.67	31.5 MWh	94.10%	2
Block 43	90.40%	-1.61	30.8 MWh	95.49%	3
Block 44	90.41%	-1.60	30.4 MWh	95.98%	0
Block 40	90.43%	-1.56	30.8 MWh	95.77%	0
Block 46	90.44%	-1.52	30.4 MWh	95.95%	5
Block 33	90.47%	-1.46	30.3 MWh	94.29%	11
Block 47	90.51%	-1.37	30.7 MWh	95.33%	0
Block 42	90.52%	-1.35	30.5 MWh	96.52%	0
Block 32	90.62%	-1.11	30.8 MWh	96.18%	4
Block 56	90.99%	-0.25	30.2 MWh	98.08%	1
Block 62	90.99%	-0.25	29.9 MWh	98.47%	14

5. System Operation

5.1 Summary of Faults and Alarms

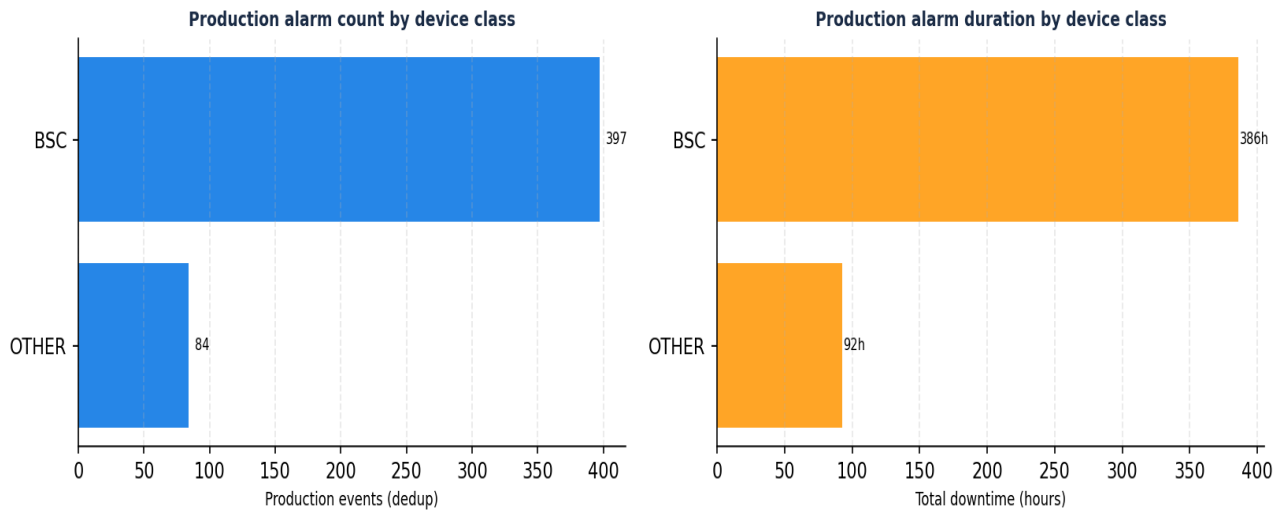
Faults (Production)

Sub-system	Reason / Cause	Occurrences	Total Hours	Blocks Affected	Resolution
BESS	Dry Node Input Fault	185	107.8 h	01, 02, 04, 05, 06, 07, 08, 09	Check field signal wiring (BMS input)

Sub-system	Reason / Cause	Occurrences	Total Hours	Blocks Affected	Resolution
BESS	Fire Alarm (Firefighting)	104	12.4 h	01, 02, 04, 05, 06, 07, 08, 09	Activate emergency response
SCC+PCS	BSC-PCS Communication	94	220.3 h	03, 06, 32, 33, 40, 41, 42, 43	Check ethernet/CAN between BSC and PCS
BESS	CMU Fault	29	3.0 h	16	Check cell monitoring unit
BESS	Insulation Impedance Low	28	3.3 h	02, 05, 06, 08, 09, 14, 25, 34	Insulation resistance test required
PCS	DC-DC Converter Fault	12	57.0 h	16, 29, 35, 64	Restart DC-DC; investigate insulation
BESS	Voltage Sample Fault	9	22.9 h	16	Replace voltage sensor; check wiring
BESS	Cell Over-voltage	7	0.7 h	16	Check BMS voltage; balance cells
BESS	Firefighting System Fault	6	15.0 h	30, 62	Inspect firefighting system
BESS	Cell Under-voltage	5	35.2 h	16	Check BMS voltage; cell health
BESS	Cell Under-voltage Protection	1	0.4 h	16	Check SOC management
PCS	DC-DC Converter Fault	1	0.4 h	16	Restart DC-DC converter

Alarms (Persistent Warnings)

Sub-system	Reason / Cause	Occurrences	Total Hours
System	LC System Alarm State	9902	37016.3 h
System	LC Fault Status	670	8562.7 h
PCS	Converter Unit Fault Status	514	539.8 h
BESS	BSC System Alarm	417	1551.2 h
BESS	LCU Other Alarm	413	128.8 h
BESS	Dry Node Input Fault	379	1414.8 h
PCS	Converter Unit Alarm Running	285	57.9 h
PPC	PPC Event	104	591.5 h
SCC+PCS	LC-PCS Communication	68	699.9 h
OTHER	Unclassified	62	1102.1 h
BESS	LCU Alarm Code	46	100.6 h
BESS	Dry Node Input Alarm	32	133.2 h
BESS	CMU Fault	6	28.3 h
SCC+BESS	LC-BSC Communication	5	0.5 h



Graph 5: Production alarm count and total downtime by device class.

5.2 Major Incidents and Breakdowns

Activated	Element	Trigger	Duration
2026-03-12 14:40	BSC 33.01.02	BSC - System Fault Status: BSC-PCS comm fault	44.3 h
2026-03-12 14:40	BSC 33.01.01	BSC - System Fault Status: BSC-PCS comm fault	44.3 h
2026-03-09 10:45	BSC 06.01.02	BSC - System Fault Status: BSC-PCS comm fault	24.3 h
2026-03-09 10:45	BSC 06.01.01	BSC - System Fault Status: BSC-PCS comm fault	24.3 h
2026-03-18 14:00	BSC 13.01.01	BSC - System Fault Status: Input dry node fault	23.6 h
2026-03-15 17:45	BSC 32.01.02	BSC - System Fault Status: BSC-PCS comm fault	19.1 h
2026-03-15 17:45	BSC 32.01.01	BSC - System Fault Status: BSC-PCS comm fault	19.1 h
2026-03-28 16:25	DC/DC 16.02.02.03	DCDC - Fault Status 2: CMU CAN output fault	17.6 h
2026-03-28 16:25	DC/DC 16.02.02.03	DCDC - Fault Status 2: CMU DI output fault	17.6 h
2026-03-28 16:25	BSC 16.02.02	BSC - CMU Fault: Cell undervoltage fault	17.6 h

6. Site Operation and Facility Management

Incidents affecting safety

No safety incidents in the reporting period.

Site visits and inspections

No site visits recorded.

7. Conclusions and Recommendations

1. Fleet RTE was **93.03%** (above the 85% target). Container-level availability was **97.14%**.
2. Across 70 blocks the fleet delivered **30,867.7 MWh** of discharge on **33,179.3 MWh** of charge, equivalent to **3067 EFC** (avg 43.8 per block).

- 3. **481** production-impacting events recorded after deduplication (plus 12,903 persistent warnings, 342 transient events filtered).
- 4. **19** block(s) flagged as underperforming vs fleet mean (z-score < -1).
- 5. **2.3** day(s) per block flagged as EXCLUDED and removed from KPI calculations.

Recommendations & Mitigation Strategies

- Continue daily monitoring of block 45 family — chronic underperformer in prior reports.
- Review BSC Input Dry Node Fault occurrences — investigate field wiring on affected blocks.
- Address Firefighting system fire alarms (104 events) — likely false positives, calibrate detectors.

Planned Activities for Next Reporting Period

- PM activities to be scheduled with O&M; team.
- Continue daily performance monitoring; escalate availability anomalies.

8. List of Annexes

- Annex 1: Daily Block KPI table — full month per block (CSV export)
- Annex 2: Alarm classification lookup applied to this report
- Annex 3: Quality-flagged days (EXCLUDED / INCOMPLETE / ESTIMATED)
- Annex 4: Block performance anomaly z-scores